

The Real Clifford Layer of Barnes–Wall Symmetry

Coset reduction, code invariants, and explicit normalizer checks

Computational verification note

17 July 2026

Abstract

We prove the exact reduction of the real Clifford sigma–MGF to coset averages over $O^+(2m, 2)$ and state precisely what the self-dual-code theorem does and does not imply for mixed coefficients. Exhaustive matrix closures in dimensions 2 and 4 recover group orders 16 and 2304, quotient orders 2 and 72, lift independence, and 16 direct coefficient matches.

1 Exact quotient formula

Let $E_m \triangleleft \mathcal{C}_m = N_{O(V_m)}(E_m)$, with

$$\mathcal{C}_m/E_m \cong O^+(2m, 2).$$

For A in the quotient and any lift $\tilde{A}$, define

$$\Phi_m(A; \mathbf{t}) = \frac{1}{|E_m|} \sum_{e \in E_m} \prod_j \det(I - t_j e \tilde{A})^{-1}.$$

Replacing $\tilde{A}$ by $e_0 \tilde{A}$ permutes E_m , so Φ_m is well defined. Partitioning $\mathcal{C}_m$ into right cosets proves

$$\mathfrak{M}_{\mathcal{C}_m, m}(\mathbf{t}) = \frac{1}{|O^+(2m, 2)|} \sum_{A \in O^+(2m, 2)} \Phi_m(A; \mathbf{t}). \quad (1)$$

Coefficient extraction gives

$$[m_\tau] \mathfrak{M}_{\mathcal{C}_m, m} = \dim \left(\bigotimes_i \text{Sym}^{\tau_i} V_m \right)^{\mathcal{C}_m}.$$

2 One-row code theorem and its boundary

For a binary self-dual code $C \leq \mathbb{F}_2^{2k}$, its genus- m complete weight enumerator is

$$\text{cwe}_m(C) = \sum_{(c^{(1)}, \dots, c^{(m)}) \in C^m} \prod_{j=1}^{2k} x_{(c_j^{(1)}, \dots, c_j^{(m)})}.$$

The Clifford invariant theorem identifies the degree- $2k$ invariant space with the span of these enumerators. In the stable range $m \geq k - 1$, they are independent and therefore

$$[m_{(2k)}] \mathfrak{M}_{\mathcal{C}_m, m} = \#\{\text{binary self-dual codes of length } 2k\} / \cong. \quad (2)$$

Consequently the stable values in degrees 2, 4, 6, 8 are 1, 1, 1, 2. The identification with genus- m complete weight enumerators and their independence in this range are external inputs from the Nebe–Rains–Sloane

invariant theorem cited below. This note proves the coset formula and verifies low-rank examples; it does not reprove that theorem. The degree-eight value is guaranteed by (2) only for $m \geq 3$; our smaller closures also happen to give two, but we report that as a direct low-rank computation rather than misuse the stable-range hypothesis.

For a general partition τ , block-symmetrizing the full code tensor produces explicit invariants. The ordinary invariant-ring theorem does not by itself show that these span every mixed fixed space. Thus (1) is exact for all τ , while a code-indexed equality for all mixed τ would require a separate multigraded theorem.

3 Explicit matrices and exhaustive checks

We generate the real Pauli matrices together with tensor placements of

$$H = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and, for $m = 2$, CNOT. Breadth-first closure yields

$$|\mathcal{C}_1| = 16 = 8 \cdot 2, \quad |\mathcal{C}_2| = 2304 = 32 \cdot 72.$$

Every closed matrix is checked to be real orthogonal. Conjugation by every selected coset representative sends the Pauli group to itself. The program then partitions the closed group into Pauli cosets, averages each coset, and compares (1) with an independent direct average. Each lift is also multiplied by a varying Pauli matrix, and every corresponding pair of coset contributions is compared before the quotient average is taken.

m	$ \mathcal{C}_m $	$ O^+ $	τ	direct	coset
1	16	2	(2)	1	1
1	16	2	(8)	2	2
1	16	2	(2, 2)	2	2
2	2304	72	(4)	1	1
2	2304	72	(8)	2	2
2	2304	72	(4, 2)	2	2

The full sweep tests eight partitions at each rank. Stable-code references are applied only when $m \geq k - 1$; all 16 rows pass.

4 Reproduction and references

```
.venv/bin/python for_this_guy/barnes_wall_lattice_automorphism_groups/
  real_clifford/verification.py
.venv/bin/marimo edit for_this_guy/barnes_wall_lattice_automorphism_groups/
  real_clifford/notebook.py
```

Join wrapped commands onto one line. For the invariant theorem, see G. Nebe, E. Rains, and N. J. A. Sloane, *The invariants of the Clifford groups*, <https://arxiv.org/abs/math/0001038>.